



CHEMISTRY
Paper 3 Free Response

9729/03
14th September 2018
2 hours

Candidates answer on separate paper.

Additional Materials: Answer Paper
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number, name and civics group on all the work you hand in.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **16** printed pages.

Section A

Answer **all** the questions in this section.

- 1 (a) There are several stages in the industrial production of methanol from methane.

The first stage involves a gaseous equilibrium between the reactants (methane and steam), and some gaseous products. Figures 1.1 and 1.2 show the percentage conversion of methane into the gaseous products under different conditions at equilibrium.

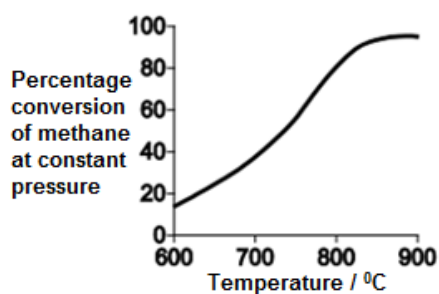


Figure 1.1

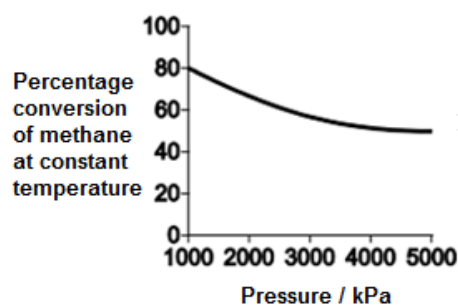


Figure 1.2

- (i) Use the information from the graphs above to deduce the sign of the

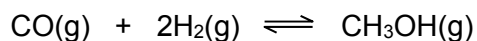
- enthalpy change, ΔH and
- entropy change, ΔS

for the first stage industrial conversion reaction of methane and steam into the gaseous products. [2]

- (ii) The optimum temperature for the industrial conversion of methane and steam into the gaseous products is between 780-880°C.

Explain why this is so. [2]

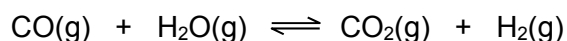
- (b) The equation shows the final stage in the production of methanol.



3.12 moles of carbon monoxide and 5.23 moles of hydrogen were placed in a sealed container. An equilibrium was established at 600 K and the total amount of gaseous molecules was found to be 7.63 moles. The total pressure was 630 kPa.

Calculate the equilibrium constant, K_p , for this reaction at 600 K and state its units. [3]

- (c) Carbon monoxide also reacts with steam.



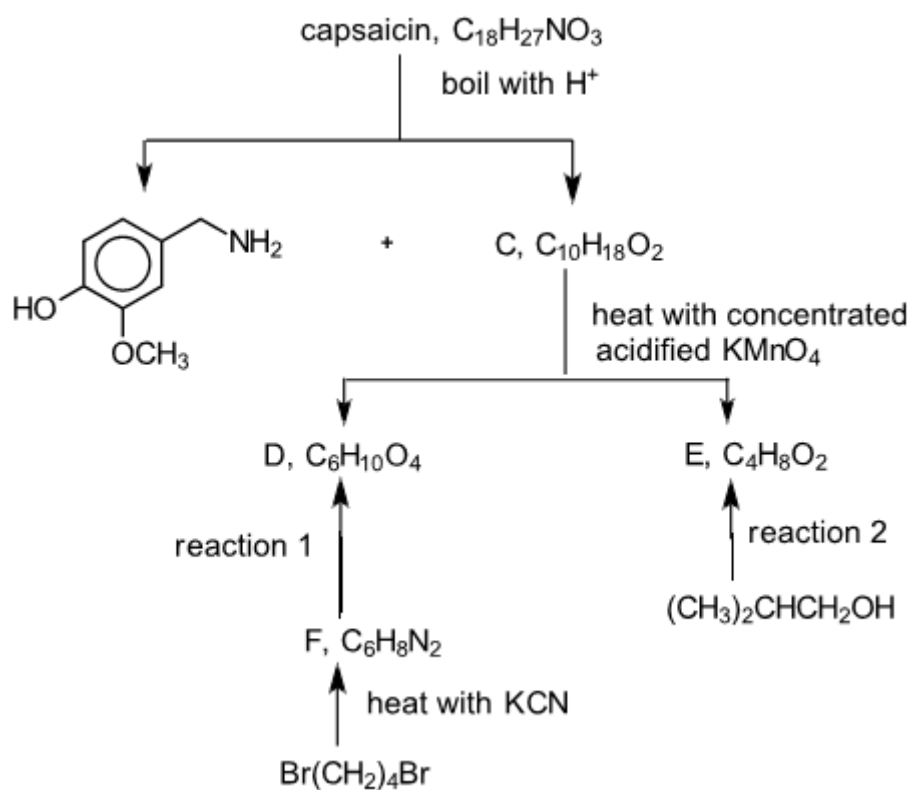
At 1100 °C, $K_c = 1.00$

In an experiment, 1 mole of carbon monoxide was mixed with 1 mole of steam, 2 moles of carbon dioxide and 2 moles of hydrogen.

Deduce, with reasons, the direction in which the reaction will shift to reach equilibrium.

[2]

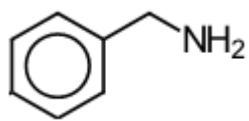
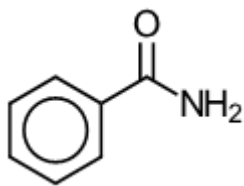
- (d) The compound responsible for the hot taste of chilli peppers is capsaicin. Its molecular structure can be deduced by the following reaction scheme.



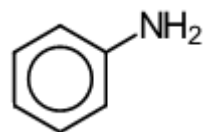
Capsaicin reacts with sodium metal and compounds C, D and E react with $\text{Na}_2\text{CO}_3(\text{aq})$.

- (i) Suggest the reagents and conditions for reactions 1 and 2. [2]
- (ii) Draw the structural formulae for the compounds C, D, E, F and capsaicin in the above reaction scheme. [5]

(e)

**X****Y**

and

**Z**

X, **Y** and **Z** are nitrogen containing compounds. State and explain the relative basicities of these 3 compounds. [3]

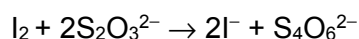
[Total:19]

- 2 This question is about nitrogen compounds and their roles in organic synthesis.

Azide is the anion with the formula N_3^- . Azide is used as a chemical preservative in hospitals and laboratories.

The azide ion in sodium azide (NaN_3) can be oxidised by iodine to form nitrogen gas. To determine the amount of sodium azide in an impure sample, the azide present is first reacted with excess iodine.

The amount of unreacted iodine is then titrated with standard sodium thiosulfate solution:



0.120 g of an impure sample of sodium azide was dissolved in water. The mixture was reacted with 25.0 cm³ of 0.050 mol dm⁻³ of aqueous iodine. The excess iodine was found to require 23.10 cm³ of 0.040 mol dm⁻³ aqueous sodium thiosulfate for reaction.

- (a) (i) Write the equation for the azide ion reacting with iodine. [1]

- (ii) Calculate the percentage purity of sodium azide in the sample. [3]

- (b) A student drew 4 structures **A** to **D** of the linear azide ion in Fig 2.1. His teacher commented one of the structures drawn was incorrect. Determine which is the **incorrect** structure, explaining your answer. [1]

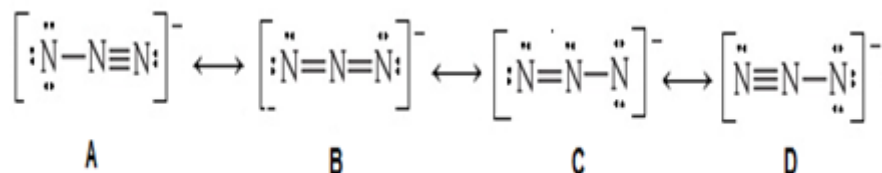


Fig. 2.1

- (c) Azides, being very versatile, can play important roles in organic reactions. An azide ion is called a pseudohalide because it has some properties which are similar to those of chloride ions. As such organic azides, RN_3 have similar chemical properties as chloroalkanes.

One organic reaction scheme, involving sodium azide to form amines is shown in Fig 2.2.

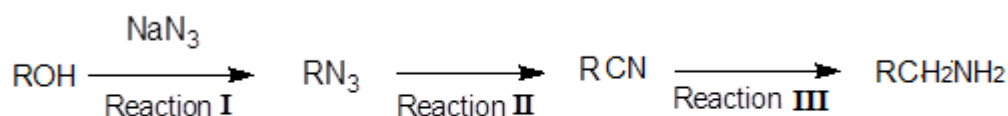
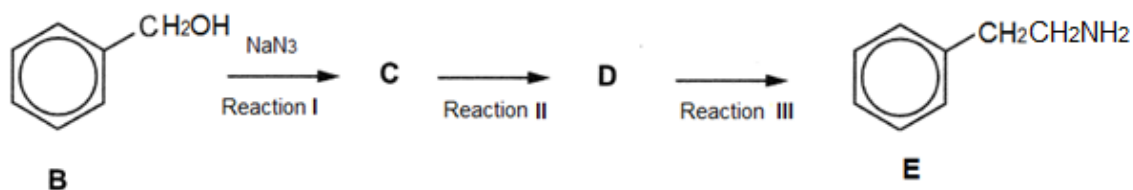


Fig. 2.2

- (i) Identify the role of azide ion in Reaction I. [1]

The 3-stage synthesis of compound **E** from compound **B**, makes use of the reaction scheme shown in Fig. 2.2.



(ii) Suggest structures for intermediates **C** and **D**, and reagents and conditions for reactions II and III. [3]

(iii) State the *type of reaction* that occur during reaction III. [1]

A reaction scheme involving compound **B** is shown in Fig.2.3.

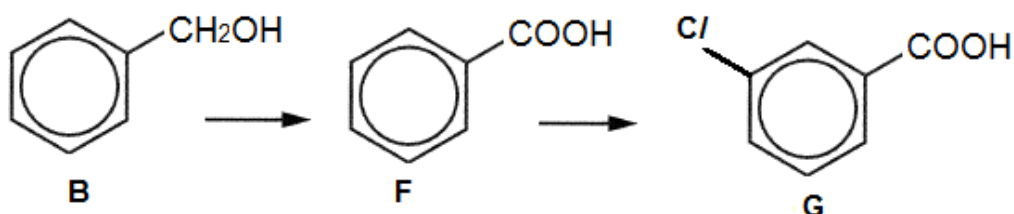


Fig. 2.3

(iv) Suggest the reagents and conditions for the conversion of **B** to **F** and for **F** to **G**. [2]

(v) Describe and explain how the acidity of **G** would compare to that of **F**. [2]

(d) Hydrogen azide, HN_3 has a boiling point of 37°C and ammonia, NH_3 has a boiling point of -33°C .

(i) Suggest a reason for the higher boiling point of hydrogen azide compared to ammonia. [1]

(ii) Explain why hydrogen azide is highly soluble in water. [1]

(e) Nitride is a compound of nitrogen where nitrogen has a oxidation state of -3. Nitrides are a large class of compounds with a wide range of properties and applications.

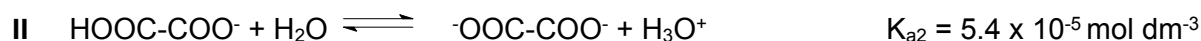
(i) State the electronic configuration of the nitride ion, N^{3-} . [1]

(ii) When a beam of nitride ions, N^{3-} was passed through an electric field, it deflected 63° towards the positive plate. Under identical conditions, a beam of azide ions, N_3^- was passed through the electric field. Determine the angle of deflection of the azide ion beam. [1]

[Total: 18]

- 3 Oxalic acid is an organic compound with the formula $\text{H}_2\text{C}_2\text{O}_4$. This colourless solid is a dicarboxylic acid. In terms of acid strength, it is about 3,000 times stronger than ethanoic acid. Its conjugate base, known as oxalate ($\text{C}_2\text{O}_4^{2-}$), is a reducing agent as well as a bidentate ligand for metal cations.

Oxalic acid dissociates in water according to the following equations.

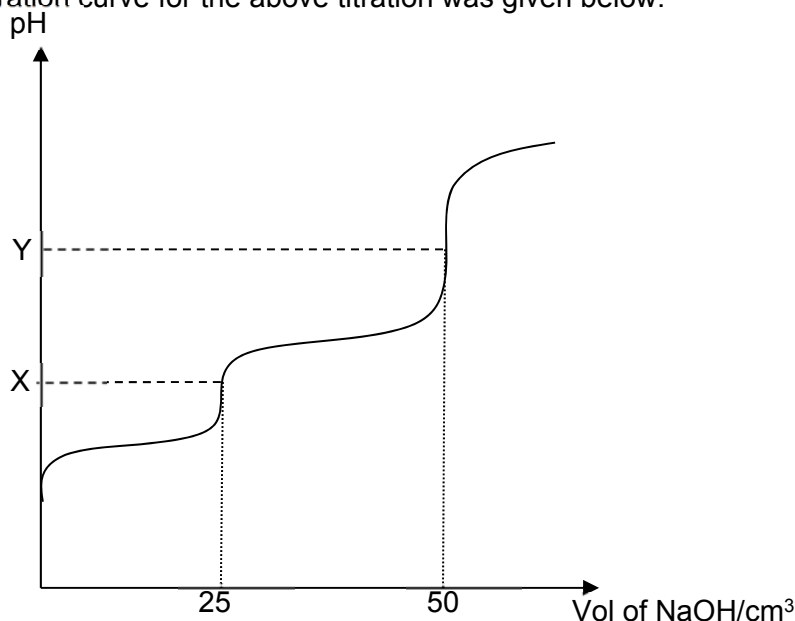


- (a) (i) Explain why the value of K_{a1} is larger than K_{a2} . [1]

- (ii) Write expressions for acid dissociation constants for equation I and II above. [1]

- (b) A 25 cm^3 sample of oxalic acid of concentration $0.100 \text{ mol dm}^{-3}$ was titrated with sodium hydroxide of concentration $0.100 \text{ mol dm}^{-3}$.

The titration curve for the above titration was given below.



- (i) State the major organic species present at points X and Y. In your answer, include the equation for the reaction that occurred to produce **each** of these two species. [2]

- (ii) It has been proven that the $[H_3O^+]$ at the first equivalent point, **X**, is given by the expression,

$$[H_3O^+] = \sqrt{K_{a1}K_{a2}}$$

Use the given expression to determine the pH value at point **X**. [1]

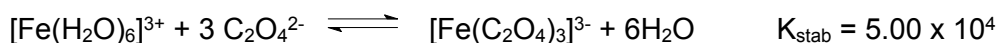
- (iii) Calculate the pH value at the second equivalent point, **Y**, given that the $[OH^-]$ can be assumed to be entirely due to the hydrolysis of the organic species at **Y**. [2]

- (c) Oxalate ions, $C_2O_4^{2-}$, are toxic to the human body. If sufficient amounts are ingested and released into the bloodstream, the high concentration of oxalate ions would cause the precipitation of calcium oxalate in the urine, which can accumulate into painful kidney stones.

Assuming that the body excretes an average of 250 mg of calcium ions per litre of urine, what is the minimum concentration of oxalate ions needed to precipitate calcium oxalate in the urine?

Numerical value of solubility product of calcium oxalate, $CaC_2O_4 = 2.7 \times 10^{-9}$. [2]

- (d) Oxalate ions, $C_2O_4^{2-}$, is also able to combine chemically with certain metals commonly found in the human body. It is able to bond chemically, behaving as a *bidentate* ligand, to transition elements, such as iron to form a complex ion.



- (i) What do you understand by the term *bidentate* ligand. [1]

- (ii) Suggest a reason why the stability constant, K_{stab} of the above is greater than 1. [1]

- (iii) Draw the structure of the complex ion, $[Fe(C_2O_4)_3]^{3-}$. Hence suggest the shape for this complex. [2]

- (e) Oxalic acid was one of the products formed when an aromatic organic compound, **A**, with molecular formula $C_{10}H_{10}O_2$ undergoes oxidation with acidified manganate(VII) to form another organic product, **B**, with the molecular formula $C_8H_8O_2$. No other organic compound was formed in the oxidation. Compound **B** reacts readily with 2 moles of $Br_2(aq)$ to form compound **E**, $C_8H_6O_2Br_2$. Compounds **A** and **B** are both soluble in NaOH and both **A** and **B** reacts with 2,4-dinitrophenylhydrazine. Compound **A** reacts with acidified dichromate to give an acid, **C**, $C_{10}H_{10}O_3$. Compound **C** reacts with $SOCl_2$ to form a sweet-smelling compound **D**, $C_{10}H_8O_2$.

Suggest structures for **A**, **B**, **C**, **D** and **E** and explain the reactions described. [10]

[Total: 23]

Turn over for Section B

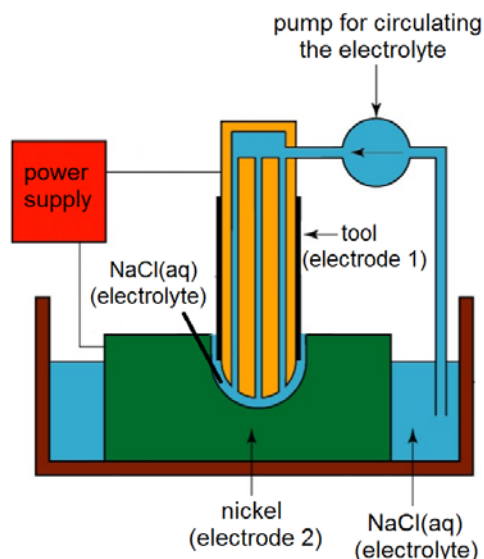
Section B

Answer **one** question from this section.

- 4 (a) Explain, with reference to relevant $E^\ominus$ values, the industrial process of the electrolytic purification of a copper alloy consisting of nickel and silver impurities. [4]

- (b) Electrochemical machining is a method used to cut small holes or cavities in metals such as nickel.

The diagram below illustrates the process. The tool (electrode 1) and nickel (electrode 2) are connected to a power supply. The electrodes are in contact with an electrolyte, NaCl(aq), which is being circulated using a pump. Some nickel is then removed by an electrochemical process when the power supply is switched on.



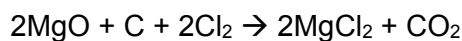
- (i) State the polarity of each electrode. Write equation for the reaction occurring at each electrode. [2]
- (ii) As the process takes place, a green solid is observed in the electrolyte. Suggest the identity of the solid. [1]
- (c) Magnesium is a Group 2 element that occurs naturally only in combination with other elements. It is the third most abundant element present in seawater, after sodium and chlorine.
- (i) Describe and explain, using $E^\ominus$ values, the relative reactivity of Group 2 elements as reducing agents. [2]

In China, magnesium is produced by electrolysis of molten magnesium chloride. Anhydrous magnesium chloride is fed continuously into electrolytic cells which are hot enough to melt it. Magnesium is produced at the cathode.

- (ii) Determine the mass of magnesium produced when a current of 40.0 A was passed through one such electrolytic cell for 10 hours. [2]

The magnesium chloride raw material needed for the electrolysis is obtained from seawater or magnesium chloride-rich brine.

If seawater is used, it is first treated to produce magnesium hydroxide which is then converted to the oxide. The oxide is then heated with carbon in a stream of chlorine at a high temperature as shown in the equation below.



- (iii) Identify the oxidising and reducing agents in the above reaction. [1]

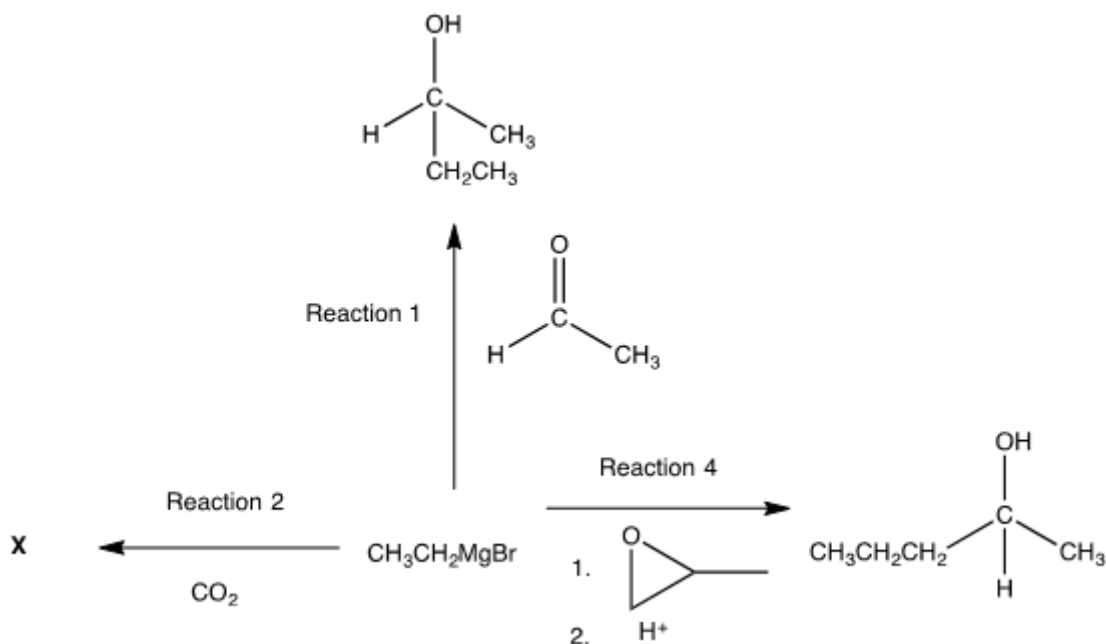
On the other hand, if magnesium chloride-rich brine is used, the solution is treated to remove impurities and concentrated by removing water. Dehydration has to be carried out in the presence of hydrogen chloride gas to avoid the reaction of magnesium chloride with water.

- (iv) Describe and explain the reaction of magnesium chloride with water. Write equation where appropriate. [2]

Grignard reagents can be formed from the reaction of magnesium metal with an alkyl or aryl bromide as shown below.



The following reaction scheme shows some reactions involving the Grignard reagent, $\text{CH}_3\text{CH}_2\text{MgBr}$. The Grignard reagent produces the carbanion, $^-\text{CH}_2\text{CH}_3$, as the reacting species.



(v) Suggest the type(s) of reaction that occur in reaction 1 and hence explain why the product mixture obtained is optically inactive. [3]

(vi) Reaction 2 occurs in a similar manner to reaction 1. Suggest the structural formula of compound **X**. [1]

(vii) Reaction 3 occurs in two stages.

Stage 1 is a bimolecular nucleophilic substitution reaction.
Stage 2 is a protonation.

Draw the mechanism for stage 1.

[2]

[Total: 20]

5 This question is about the chemistry of iron-containing compounds.

(a) Iron is the cheapest and one of the most abundant of all metals, comprising nearly 5.6% of the earth's crust and nearly the earth's entire core. It exists in a wide range of oxidation states and undergo redox reactions readily.

(i) By means of a fully labelled diagram, describe how the standard electrode potential of the $\text{Fe}^{3+}/\text{Fe}^{2+}$ system can be measured. Indicate the direction of electron flow. [3]

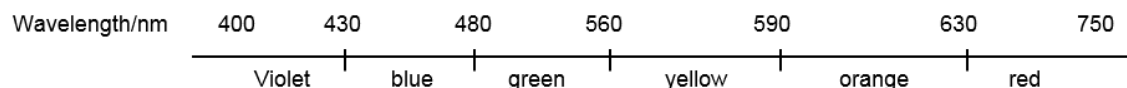
In Experiment 1, when aqueous iron(III) sulfate is boiled with an excess of potassium cyanide, an orange-red solution is obtained. The solution remains orange-red on addition of aqueous potassium iodide.

In Experiment 2, when aqueous iron(III) sulfate is added to aqueous potassium iodide, a brown solution is obtained.

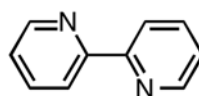
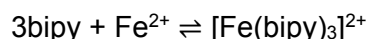
(ii) State the type of reaction that occurred in each experiment. Write a balanced equation for any reaction that occurred. [3]

(iii) With reference to relevant E° values, explain the observation in Experiment 1 upon addition of aqueous potassium iodide. [1]

The colour spectrum of the visible region of the electromagnetic spectrum is as shown:



Iron(II) ions form an intense red complex with 2,2'-bipyridine (bipy) according to the following equation:



2,2'-bipyridine

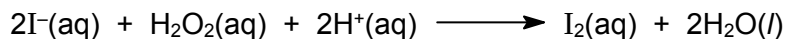
The presence of ligands in complexes causes a splitting of the energy of d orbitals into two groups with an energy gap, ΔE , between them.

(iv) Given that aqueous iron(II) ion is green in colour, state and explain which ligand, water or 2,2'-bipyridine, results in a larger ΔE . [2]

(v) Draw a labelled diagram of one of the d orbitals at the **higher** energy level in $[\text{Fe}(\text{bipy})_3]^{2+}$. [1]

- (b) In the laboratory, iron(II) ions function as a catalyst for the reaction of iodide with hydrogen peroxide.

In an experiment to investigate the kinetics of the reaction, hydrogen peroxide is added to a solution of potassium iodide in the absence of the iron(II) catalyst at 25 °C, and the iodide ions are oxidised according to the equation:



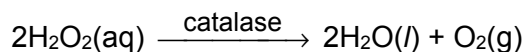
The following data was obtained.

Expt	Initial $[\text{I}^-]$ / mol dm^{-3}	Initial $[\text{H}_2\text{O}_2]$ / mol dm^{-3}	Initial $[\text{H}^+]$ / mol dm^{-3}	Initial rate of reaction / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.01	0.01	0.0005	1.15×10^{-6}
2	0.01	0.02	0.0005	2.30×10^{-6}
3	0.02	0.01	0.0005	2.30×10^{-6}
4	0.01	0.01	0.001	1.15×10^{-6}

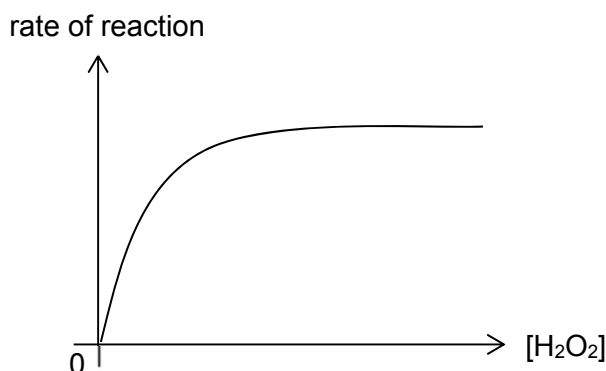
- (i) Use the data above to deduce the order of reaction with respect to each of the reagents, showing how you arrive at your answers. Hence, write the rate equation for the above reaction. [3]
- (ii) Calculate a value for the rate constant. Include units in your answer. [1]
- (iii) Explain, with the aid of a suitable diagram, why the rate of decomposition of hydrogen peroxide is expected to proceed faster at 35 °C. [2]

- (c) Some protein-built enzymes form complexes with the metal ions at their active site. An example is catalase, which is an enzyme in the liver that breaks down harmful hydrogen peroxide into oxygen and water. It contains four iron-containing heme groups that allow the enzyme to react with the hydrogen peroxide.

The decomposition of hydrogen peroxide solution to form water and oxygen gas is as shown:



- (i) What do you understand by the term *enzyme*? [1]
- (ii) A series of experiments were carried out to study the effect of hydrogen peroxide concentration on the rate of reaction, by using different concentrations of hydrogen peroxide at 25 °C. The following graph was obtained.



Account for the shape of the graph.

[3]

[Total: 20]